

Incident Report Analysis

DoS Attack — ICMP Flood via Unconfigured Firewall

Prepared by: G S Krishna Chandra | Framework: NIST CSF | Course: Google Cybersecurity Certificate — Course 3 (Connect and Protect: Networks and Network Security)

Scenario

A multimedia company offering web design, graphic design, and social media marketing services experienced a Denial of Service (DoS) attack that disrupted its internal network for approximately two hours. A malicious actor exploited an unconfigured firewall to flood the network with ICMP ping packets, preventing all normal internal traffic from accessing network resources. The incident management team responded by blocking ICMP packets, taking non-critical services offline, and restoring critical services. Post-incident measures included firewall rate limiting, source IP verification, network monitoring software, and IDS/IPS deployment.

NIST CSF Incident Analysis

Summary	The organization experienced a DoS (ICMP flood) attack that took down internal network services for two hours. A malicious actor exploited an unconfigured firewall to send a massive volume of ICMP echo requests, overwhelming network capacity. Normal internal traffic could not access any resources during the attack. The team responded by blocking ICMP packets, taking non-critical services offline, and restoring critical services. Post-incident controls were implemented including firewall rate limiting, IP spoofing verification, network monitoring, and IDS/IPS.
Identify	Attack type: DoS — ICMP Flood (Ping Flood). The malicious actor exploited an unconfigured firewall — a critical security gap — to send unrestricted ICMP traffic into the network. Affected systems: entire internal network infrastructure, all networked devices, and all services dependent on network connectivity. Root cause: firewall misconfiguration that allowed unrestricted inbound ICMP packets without rate limiting or source IP verification.
Protect	Protections implemented and recommended: (1) Firewall rule added to rate-limit incoming ICMP packets; (2) Source IP address verification on the firewall to detect and block spoofed IPs; (3) Regular firewall configuration audits to identify misconfigurations proactively; (4) Formal firewall policy defining rules for all inbound/outbound traffic types; (5) Staff awareness training on recognising and reporting abnormal network behaviour.

Detect	Detection capabilities implemented and recommended: (1) IDS/IPS deployed to monitor for suspicious traffic including abnormal ICMP volumes; (2) Network monitoring software configured to establish traffic baselines and alert on anomalies; (3) Firewall logging enabled to capture all inbound and outbound traffic; (4) SIEM tool to aggregate logs from firewall, IDS/IPS, and network devices, giving the security team centralised real-time visibility into potential threats.
---------------	--

Respond	Response plan for future incidents: (1) Immediately isolate affected network segments to contain the attack; (2) Apply rate limiting and IP blocking rules at the firewall; (3) Take non-critical services offline to preserve bandwidth for critical systems; (4) Notify management and stakeholders of the incident and estimated impact; (5) Document all actions taken during the incident for post-event analysis; (6) Analyse firewall and IDS logs to identify attack source, entry point, and timeline; (7) Review and update incident response plan after each event.
Recover	Recovery steps: (1) Restore critical network services first, followed by non-critical services once the attack is resolved; (2) Verify integrity of all networked devices post-attack to rule out secondary compromise; (3) Review and restore any data or configurations altered during the incident; (4) Conduct a post-incident review to document lessons learned; (5) Update firewall rules, IDS/IPS signatures, and monitoring thresholds based on observed attack patterns; (6) Communicate recovery status and timelines to affected staff and management.
Reflections / Notes	This incident highlights the critical importance of firewall configuration management. A single misconfiguration allowed a straightforward DoS attack to disrupt the entire network for two hours. The reactive measures implemented post-attack — rate limiting, IP verification, IDS/IPS — should have been baseline configurations from the start. Going forward, firewall audits should be scheduled regularly and all firewall rule changes should go through a formal change management review process to prevent similar incidents.